Week 7 Report – Baseline vs. Hyperparameter-Optimized Classifiers

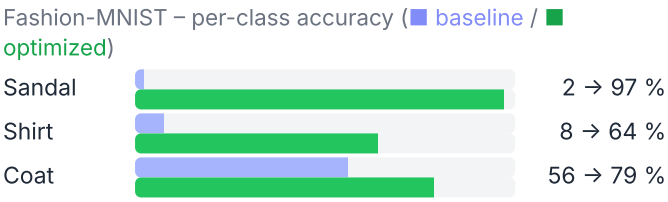
EE 471 · Özgür Şentürk · 9 November 2026 · PyTorch 2.15 on an RTX 5070 Ti Laptop GPU · code: github.com/ozgursntrk/EE471 (Week7)

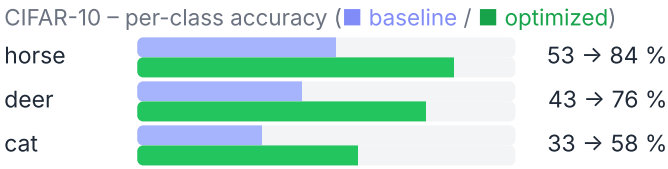
1. Setup

Fashion-MNIST (28×28 grayscale, 10 garments) uses the fully connected network of the PyTorch Quickstart (784-512-512-10). **CIFAR-10** (32×32 colour, 10 objects) uses the LeNet-style CNN of the "Training a Classifier" tutorial, with tunable widths. The **baseline** models use exactly the tutorial hyperparameters. The **optimized** models come from an **Optuna** search (TPE sampler, 20 trials, median pruning) over optimizer (SGD/Adam), learning rate, momentum, weight decay, dropout, batch size, epochs and network width. Every trial was scored on a validation split of the training set (50k/10k and 45k/5k), so the 10k test set was used only once, for the final models, which were retrained on the full training set.

2. Results (test set, 10 000 images)

Model	Hyperparameters	Parameters	Test accuracy
Fashion baseline	SGD, lr 0.001, momentum 0, batch 64, 5 epochs	670 k	65.04 %
Fashion optimized	SGD, lr 0.044, momentum 0.56, weight decay 1e-5, batch 32, 15 epochs	670 k	**88.94 % (+23.9)**
CIFAR-10 baseline	SGD, lr 0.001, momentum 0.9, batch 4, 2 epochs, 6/16 conv channels	62 k	56.71 %
CIFAR-10 optimized	Adam, lr 0.00022, weight decay 1e-4, dropout 0.3, batch 32, 15 epochs, 48/128 channels	553 k	**74.23 % (+17.5)**





3. Why hyperparameter optimization matters

- **Same network, +24 points.** The Fashion models have the identical architecture (670 k parameters). Only the training settings differ. The baseline is under-trained: its learning rate is too small and its loss was still falling after 5 epochs. It almost never predicts "Sandal" (2 %) or "Shirt" (8 %). With a 44× larger learning rate, momentum and more epochs, the same network reaches 88.9 %.
- **Bad settings can destroy a model.** Several trials stayed at ~10 % validation accuracy, which is random guessing for 10 classes (Adam with lr ≈ 0.02–0.05, or SGD with lr ≈ 0.0002). Over the 20 trials, validation accuracy ranged from 10 % to 89 % (Fashion) and from 9 % to 74 % (CIFAR-10).
- **Capacity is a hyperparameter too.** For CIFAR-10 the search chose 8× wider convolution layers (48/128 channels) plus dropout 0.3 and weight decay against over-fitting. The hardest classes improved most: cat 33 → 58 %, deer 43 → 76 %.
- **The search is cheap.** Thanks to the GPU and median pruning (7/20 and 10/20 trials stopped after 2 epochs), each search took only about 5–6 minutes, a small cost for a 17–24 point gain.

4. Limitations

Both datasets are small and low-resolution. On photos from the web the Fashion model is unreliable (e.g. a white sneaker on a white background was classified as "Sandal"): Fashion-MNIST items are centred, light on black and always face the same way, and a fully connected network is not translation invariant (domain shift). The remaining errors are between similar classes (shirt / T-shirt / coat, cat / dog). Data augmentation, batch normalization, a deeper CNN and a longer search would be the next steps.